# Day 3: OSI Model

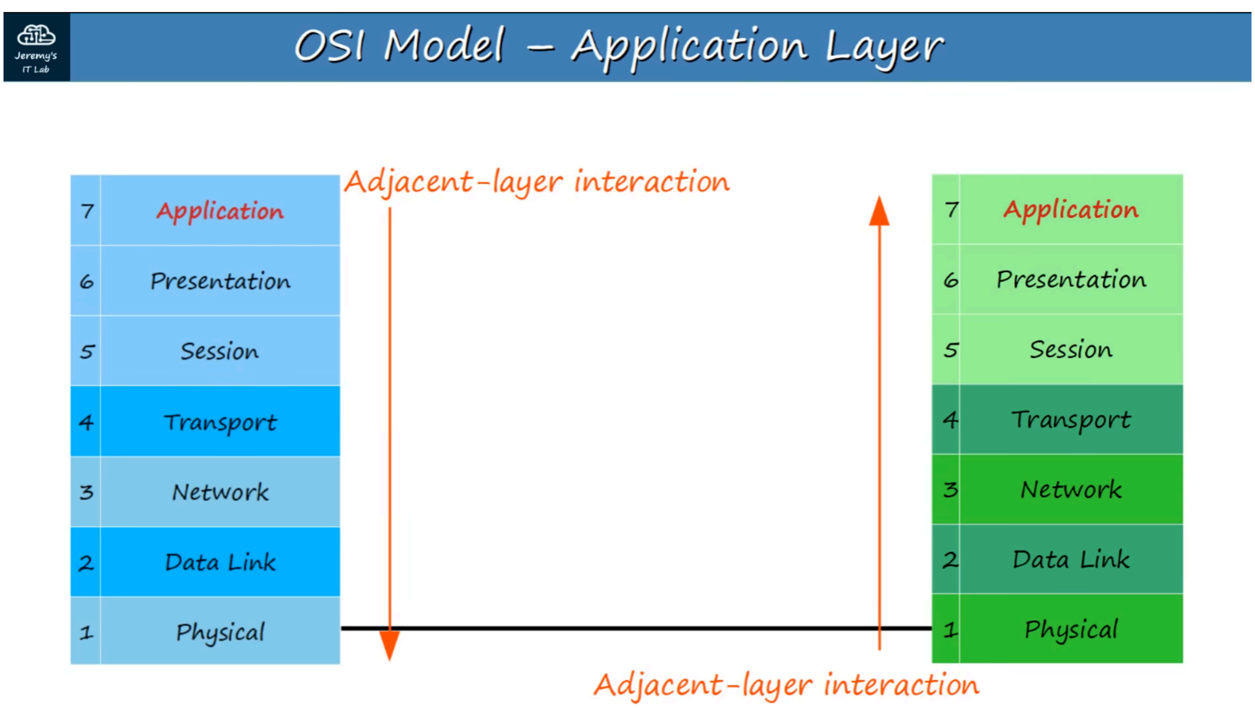
## What is a networking model?

Networking models categorize and provide a structure for networking protocols and standards.

(Protocols are a set of logical rules defining how network devices and software should work)

## OSI MODEL

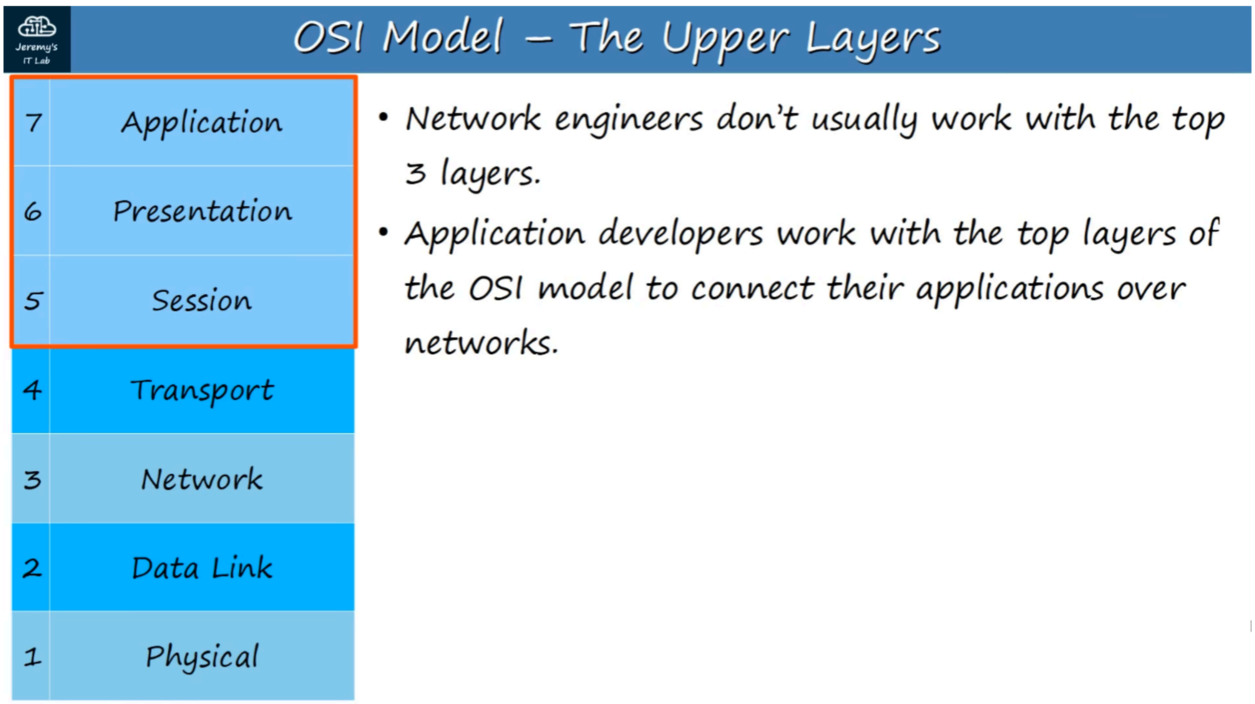
- Open Systems Interconnection Model
- Conceptual model that categorizes and standardizes the different functions in a network.
- Created by the "International Organization for Standardization" (ISO)
- Functions are divided into 7 "Layers"
- These layers work together to make the network work.



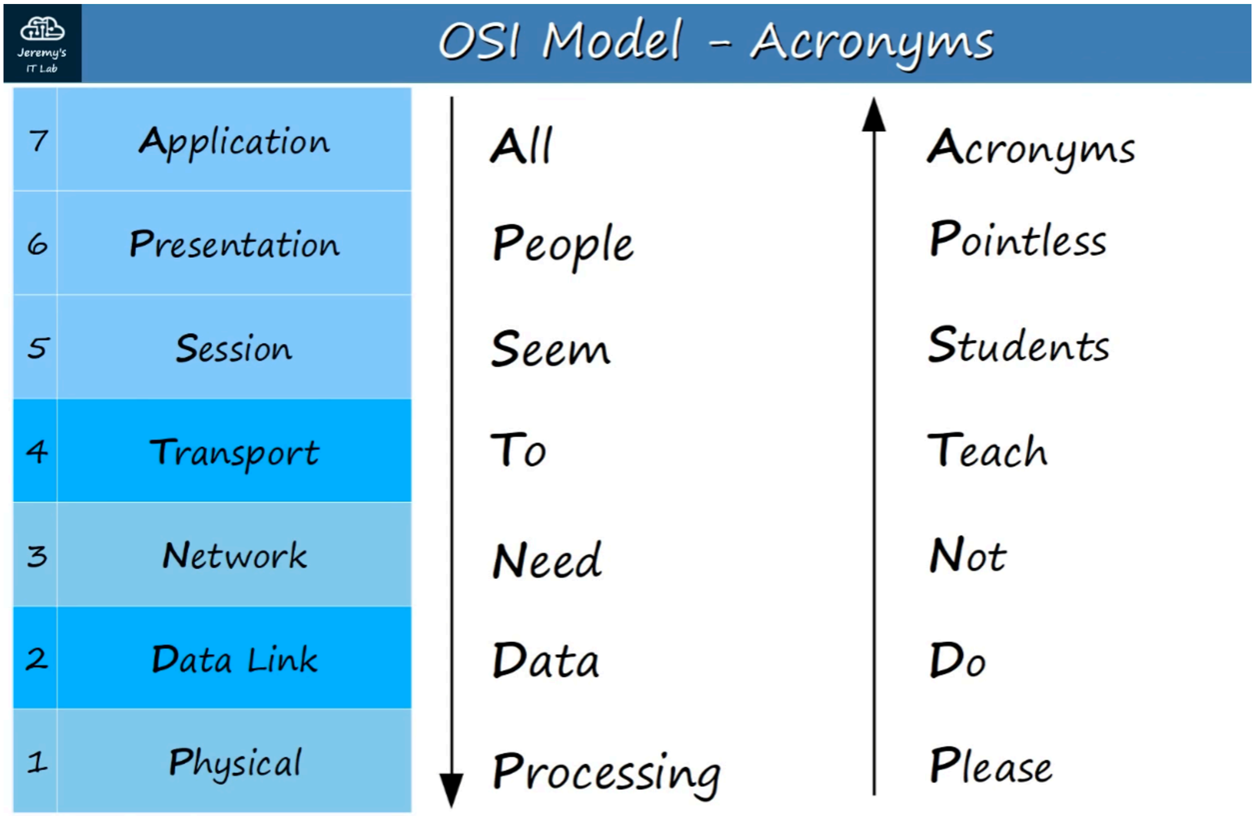
As data moves from the top layer, downward, the process is called "encapsulation"

As data moves from the bottom layer, upward, the process is called "de-encapsulation"

When interactions occur on the same layer, it's called "same-layer interaction"




Mnemonic to help remember the Data Layer Names / Order




### The layers are :

## 7 - APPLICATION

- This Layer is closest to end user.
- Interacts with software applications.
- HTTP and HTTPS are Layer 7 protocols

Functions of Layer 7 include:

- Identifying communication partners
- Synchronizing communication

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## 6 - PRESENTATION

- Translates data to the appropriate format (between Application and Network formats) to be sent over the network.

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## 5 - SESSION

- Controls dialogues (sessions) between communicating hosts.
- Establishes, manages, and terminates connections between local application and the remote application.

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Network engineers don't usually work with the top 3 layers. Application developers work with the top layers of the OSI model to connect their applications over networks.

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## 4 - TRANSPORT

- Segments and reassembles data for communication between end hosts.
- Breaks large pieces of data into smaller segments which can be more easily sent over the network and are less likely to cause transmission problems if errors occur.
- Provides HOST-TO-HOST (end to end) communication

When Data from Layer 7-5 arrives, it receives a Layer 4 Header in the Transport layer.

<< DATA + L4 Header >>

This is called a SEGMENT.

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## 3 - NETWORK

- Provides connectivity between end hosts on different networks (ie: outside of the LAN).
- Provides logical addressing (IP Addresses).
- Provides path selection between source and destination
- **ROUTERS** operate at Layer 3.

When Data and the Layer 4 Header arrive in the Network Layer, it receives a Layer 3 Header.

<< DATA + L4 Header + L3 Header >>

This is called a **PACKET**.

## 2 - DATA LINK

- Provides NODE-TO-NODE connectivity and data transfer (for example, PC to Switch, Switch to Router, Router to Router)
- Defines how data is formatted for transmission over physical medium (for example, copper UTP cables)
- Detects and (possibly) corrects Physical (Layer 1) errors.
- Uses Layer 2 addressing, separate from Layer 3 addressing.
- **SWITCHES** operate at Layer 2

When the Layer 3 Packet arrives, a Layer 2 Trailer and Header are added.

<< L2 Trailer + DATA + L4 Header + L3 Header + L2 Header >>

This is called a FRAME.

All the steps leading up to transmission is called ENCAPSULATION. When the frame is sent to the receiver, it then goes through the reverse process, DE-ENCAPSULATION, stripping off layers while travelling from OSI Layer 1 to Layer 7.

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## 1 - PHYSICAL

- Defines physical characteristics of the medium used to transfer data between devices. For example : voltage levels, maximum transmission distances, physical connectors, cable specs.
- Digital bits are converted into electrical (for wired connections) or radio (for wireless connections) signals.
- All of the information in SECTION 2 (NETWORKING DEVICES) is related to the Physical Layer

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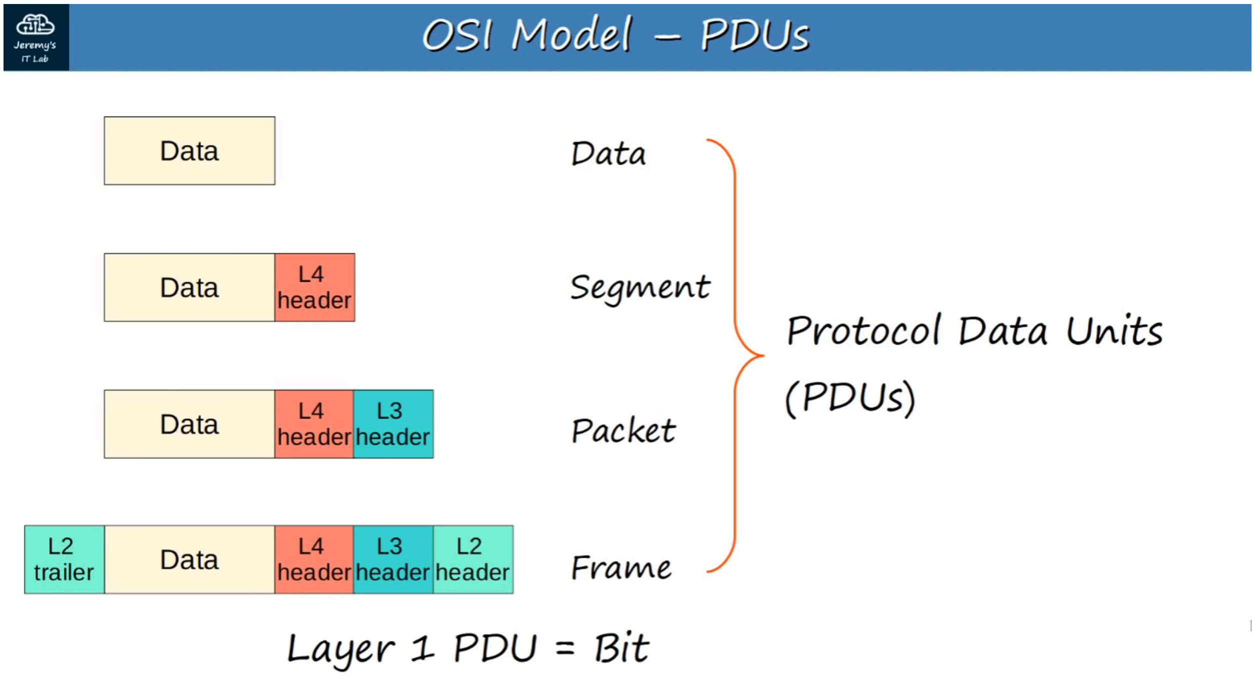
## OSI MODEL - PDU's




A PDU is a Protocol Data Unit. Each step of the process is a PDU.

| OSI LAYER # | PDU NAME | PROTOCOL DATA ADDED |
|---|---|---|
| 7-5 | DATA | Data |
| 4 | SEGMENT | Layer 4 Header Added |
| 3 | PACKET | Layer 3 Header Added |
| 2 | FRAME | Layer 2 Trailer and Header Added |
| 1 | BIT | 0s and 1s Transmission |

<< L2 Trailer + DATA + L4 Header + L3 Header + L2 Header >>

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## TCP/IP Suite

- Conceptual model and set of communications protocols used in the Internet and other networks.
- Known as TCP/IP because those are two of the foundational protocols in the suite.
- Developed by the US Dept. of Defense through DARPA (Defense Advanced Research Projects Agency).

- Similar structure to the OSI Model, but fewer layers.
- THIS is the model actually in use in modern networks.
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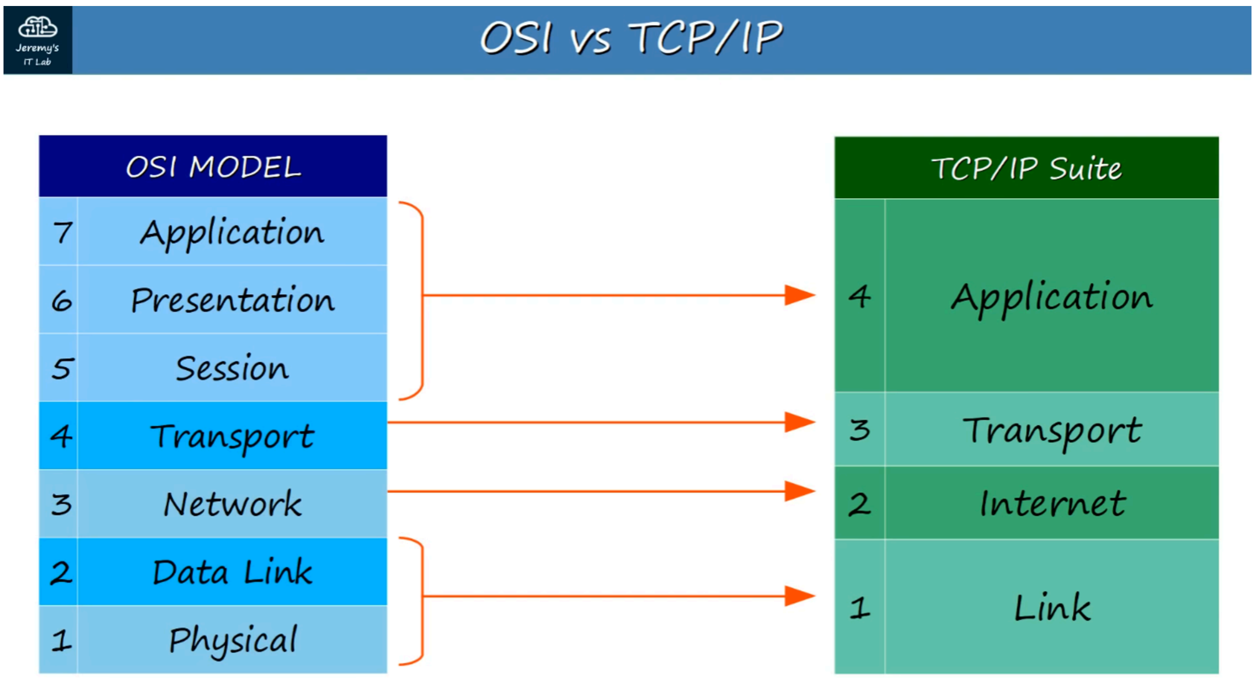
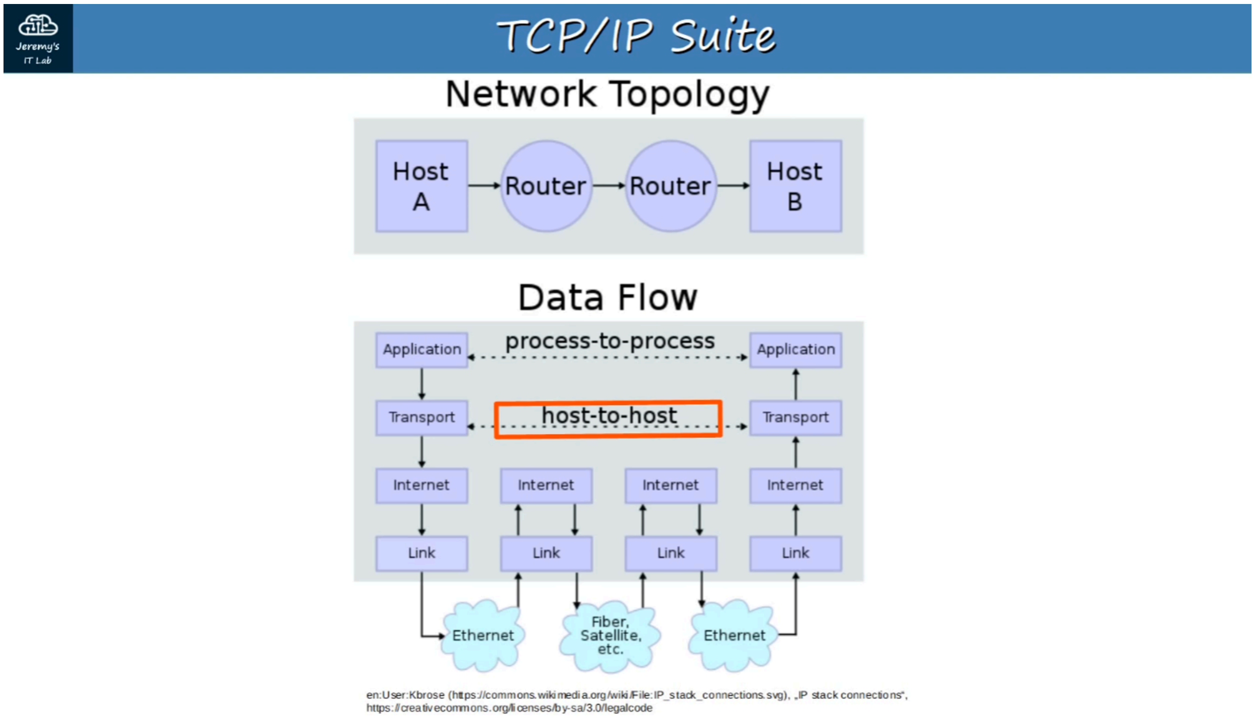
    - Note : The OSI Model still influences how network engineers think and talk about networks.




## Layer Interactions




Adjacent-Layer Interactions:

- Interactions between different layers of the OSI Model on same host.

Example:

Layers 5-7 sending Data to Layer 4, which then adds a Layer 4 header (creating a SEGMENT).

Same-Layer Interactions:

- Interactions between the same Layer on different hosts.
- The concept of Same-Layer interaction allows you to ignore the other layers involved and focus on the interactions between a single layer on different devices.

Example:

The Application Layer of YouTube's web server and your PC's browser.